Java Assignment: 1

1. What is Java?
--> Java is object oriented programming language which is Platform Dependent,

---

2. Is Java platform dependent or independent?
--> Java is Platform Dependent

---

3. What is JVM?
--> JVM stands for Java Virtual Machine which interprets bytecode and runs Java programs on any platform.

---

4. What is JDK and JRE?
--> JDK (Java Development Kit) is for developing Java apps & JRE (Java Runtime Environment) is for running them.

---

5. Why is Java called “Write Once, Run Anywhere”?
--> Java is called “Write Once, Run Anywhere” because bytecode runs on any JVM, regardless of OS.

---

6. What is a class in Java?
-->A class in Java is a blueprint for creating objects, defining attributes and methods.

---

7. What is an object?
--> An object is an instance of a class, containing data and methods to manipulate it.

---

8. What is the main method in Java?
--> The main method is the entry point where Java program execution begins.

---

9. What is the use of System.out.println()?
--> System.out.println() prints output to the console and adds a newline.

---

11. What are data types in Java?
--> Primitive datatypes and Non-primitive datatypes.

---

12. Give examples of primitive data types.
--> int, char, float, double, Boolean.

---

13. What is a variable?
--> Variable is name given to memory location.

---

14. How do you declare a variable in Java?
--> Datatype_variableName;

---

15. What is the difference between int and float?
--> int stores whole numbers; float stores decimal numbers with precision.

---

16. What is type casting?
--> Type casting converts one data type to another.

---

17. What is the default value of int?
--> Default value of int is 0.

---

18. What is a keyword in Java?
--> A keyword is a reserved word with a specific meaning.

---

19. Can we use reserved keywords as variable names?
--> No, reserved keywords cannot be used as variable names.

---

20. What is the difference between = and ==?
--> = is assignment,
== is comparison for equality.

---

21. What is an if statement?
--> if statement executes code conditionally based on a boolean expression.

---

22. Difference between if and switch?

--> if checks a boolean expression,
switch checks multiple discrete values of a variable.

---

23. What is a loop?
--> A loop repeatedly executes a block of code while a condition is true.

---

24. Types of loops in Java?
--> for, while, do-while.

---

25. Difference between for loop and while loop?
--> for loop has initialization, condition, increment in one line,
while checks condition before each iteration.

---

26. What is break statement?
--> break exits the loop or switch statement immediately from nearest block.

---

27. What is continue statement?
--> continue skips the current iteration and moves to the next loop iteration.

---

28. What will happen if we don't write the main method?
--> Compilation Error.

---

29. Can we run a Java program without class?
--> No, Java requires a class to define a program structure.

---

30. What happens if we don't compile a Java program?
--> Without compilation, bytecode isn't generated; the program won't run.

---

31. Can a variable change its data type?
--> No, a variable's type is fixed at declaration.

---

32. What is the size of int in Java?
--> 4 Bytes.

---

33. Why Java does not support pointers directly?
--> Java avoids pointers for security and simplicity, using references instead.

==========
****

1. Write a Java program to declare variables of all primitive data types and print their values.

```
--> class DatatypesDemo{
     public static void main(String args[]){
            int a = 10;
            float b = 20.5f;
            char c = 'S';
            Boolean d = true ;

            System.out.println(a);
            System.out.println(b);
            System.out.println(c);
            System.out.println(d);
        }
}
```

---

2. Take two numbers (int and double) and perform:
○ Addition
○ Division
○ Type casting result to int

```
--> class numbersDemo{
    public static void main(String args[]){
         int a = 40;
         double d = 20.0;
         double res1 = (int)(a + d);
         double res2 = (int)(a / d);

         System.out.println(res1);
         System.out.println(res2);
  }
}
```

---

3. Write a program to convert:
○ int → double
○ double → int
--> class datatypeDemo{

```
        public static void main(String args[]){
                int a = (int)458612358;
                double d = 585645;

                System.out.println(a);
                System.out.println(d);
        }

}
```

---

4. Write a program to find ASCII value of a character.
--> class Assign{

```
     public static void main(String args[]){
            int a = 'A';
            System.out.println("ASCCI of A : "+ a);
           }
}
```

---

5. Take a character input and print:
○ Next character
○ Previous character
-->

```
import java.util.Scanner;
class charIpDemo{
       public static void main(String args[]){
            char arr[] = new char[4];
            Scanner sc = new Scanner(System.in);
            System.out.println("Enter characters : ");
            for(int i=0; i<4; i++){
                 arr[i] = sc.next().charAt(0);
                  }
            System.out.println("Characters are: ");
            for(int i=0; i<4; i++){
                 System.out.print(arr[i]+"\t");
```

```
                }
            System.out.println();
        }
}
```

---

6. Write a program to check whether a number is:
○ Positive
○ Negative
○ Zero
-->

```
class Assign{
    public static void main(String args[]){
        int a = 0;
        if(a >0){
            System.out.println("The number is Positive");
        } else if(a == 0){
            System.out.println("The number is Zero");

            }else{
            System.out.println("The number is Negative");

            }
    }

}
```

---

7. Write a program to find the largest of three numbers.
-->

```
class largestNumberDemo{
   public static void main(String args[]){
        int a = 12;
        int b = 20;
        int c = 25;

       if(a>b && a>c){
           System.out.println(a+" is largest");
           }else if(b>a && b>c){
            System.out.println(b+" is largest");
           }else{
            System.out.println(c+" is largest");
           }
```

```
    }

}
```

---

8. Write a program to check whether a year is a leap year.
-->

```
class Assign{
      public static void main(String args[]){
              int year = 2024;

             if((year%4 == 0 && year % 100 != 0)|| (year % 400 == 0)){
                System.out.println("The year is leap");
                 }else{
                System.out.println(year + " The year is non-leap");

        }
    }
}
```

---

9. Create a simple calculator using switch:
○ Input: two numbers + operator
○ Perform: +, -, *, /
-->

```
import java.util.Scanner;
  class operatorDemo{

  public static void main(String args[]){
          Scanner sc= new Scanner(System.in);

           System.out.println("Enter first number: ");
           int num1 = sc.nextInt();
           System.out.println("Enter Second number: ");
           int num2 = sc.nextInt();

           System.out.println("Enter Operator : (+ , - , * , /) : ");
           char operator = sc.next().charAt(0);
           int result;

           switch(operator){
                case'+':
                      result = num1 + num2;
                      System.out.println("Addition : "+ result);
                      break;
```

```
            case'-':
                result = num1 - num2;
                System.out.println("Substraction : "+ result);
                break;
            case'*':
                result = num1 * num2;
                System.out.println("Multiplication : "+ result);
                break;
            case'/':
                if(num2 != 0){
                result = num1 / num2;
                System.out.println("Division : "+ result);
                }else{
                System.out.println("Division by zero not possible ");
                }
                break;
            default:
               System.out.println("Invalid Operator");
             }
        sc.close();
    }
}
```

---

10. Write a program to check whether a character is:

● Vowel or consonant

```
-->import java.util.Scanner;
class VowelConsonant {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a character: ");
        char ch = sc.next().charAt(0);
        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' ||
            ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U') {
            System.out.println(ch + " is a vowel.");
        } else if ((ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z')) {
            System.out.println(ch + " is a consonant.");
        } else {
            System.out.println("Invalid input;");
        }
    }
}
```

---

11. Print numbers from 1 to 50 using:
● for loop
● while loop
-->
```
class primeNumbersDemo {
    public static void main(String[] args) {
        System.out.println("Using for loop:");
        for (int i = 1; i <= 50; i++) {
            System.out.print(i+"\t");
        }
        System.out.println();
        System.out.println("Using while loop:");
        int i = 1;
        while (i <= 50) {
            System.out.print(i+ "\t");
            i++;
        }
    }
}
```

---

12. Write a program to print sum of first N natural numbers.
-->
```
import java.util.Scanner;
class sumPrimeNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Num: ");
        int Num = sc.nextInt();
        int sum = Num * (Num + 1) / 2;
        System.out.println("Sum = " + sum);
    }
}
```

---

13. Write a program to print multiplication table of a number.
-->

```
import java.util.Scanner;
class MultiplicationTable {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = sc.nextInt();
        for (int i = 1; i <= 10; i++) {
            System.out.println(num + " * " + i + " = " + (num * i));
        }
    }
}
```

---

14. Write a program to reverse a number.
-->

```
import java.util.Scanner;
class ReverseNumber {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = sc.nextInt();
        int reversed = 0;
        while (num != 0) {
            reversed = reversed * 10 + num % 10;
            num /= 10;
        }
        System.out.println("Reversed number: " + reversed);
    }
}
```

---

15. Write a program to count digits in a number.
--> import java.util.Scanner;

```
class countDigits {
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = sc.nextInt();
        int count = 0;
        while (num != 0){
            num /= 10;
            count++;
```

```
        }
        System.out.println("Number of digits: " + count);
    }
}
```

---

16. Write a program to check whether a number is palindrome.

```
-->  import java.util.Scanner;
class palindromeCheck {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = sc.nextInt();
        int original = num;
        int reversed = 0;
        while (num != 0) {
            reversed = reversed * 10 + num % 10;
            num /= 10;
        }
        System.out.println(original == reversed ? "Palindrome" : "Not Palindrome");
    }
}
```

---

17. Write a program to find factorial of a number.

```
-->import java.util.Scanner;
class factorialDemo {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = sc.nextInt();
        long f = 1;
        for (int i = 1; i <= num; i++) {
            f *= i;
        }
        System.out.println("Factorial = " + f);
    }
}
```

---

18. Write a program to print Fibonacci series up to N terms.

```
--> import java.util.Scanner;
class fibonacciSeries {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter N: ");
        int N = sc.nextInt();
        int a = 0, b = 1;
        System.out.print(a + " " + b);
        for (int i = 2; i < N; i++) {
            int next = a + b;
            System.out.print(" " + next);
            a = b;
            b = next;
        }
    }
}
```

---

19. Print:

```
*
**
***
****
--> class Assign{
    public static void main(String args[]) {
        for (int i = 1; i <= 4; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

---

20. Print:

```
1
12
123
1234
-->class firstPattern{
    public static void main(String args[]) {
        for (int i = 1; i <= 4; i++) {
```

```
            for (int j = 1; j <= i; j++) {
                System.out.print(j);
            }
            System.out.println();
        }
    }
}
```

---

21. Print:

```
****
***
**
*
```

```
--> class secondPattern{
    public static void main(String args[]) {
        for (int i = 4; i >= 1; i--) {
            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

---

22. Write a program to check whether a number is prime.

```
-->
import java.util.Scanner;
class checkPrimeNumber{
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        boolean b = true;
        for (int i = 2; i <= n/2; i++) {
            if (n % i == 0) {
                b = false;
                break;
            }
        }
        System.out.println(n + " is " + (b ? "prime" : "not prime"));
```

```
    }
}
```

---

23. Write a program to find sum of even and odd numbers separately (1 to N).
-->

```
import java.util.Scanner;
class sumDemo{
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter N: ");
        int n = sc.nextInt();
        int evenSum = 0;
        int oddSum = 0;
        for (int i = 1; i <= n; i++) {
            if (i % 2 == 0) evenSum += i;
            else oddSum += i;
        }
        System.out.println("Even sum: " + evenSum + ", Odd sum: " + oddSum);
    }
}
```

---

24. Write a program to find largest digit in a number.
-->

```
import java.util.Scanner;
class largestNumberDemo{
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        int max = 0;
        while (n > 0) {
            int d = n % 10;
            if (d > max){
                max = d;
            }
           else{
             n /= 10;
              }
        }
        System.out.println("Largest digit: " + max);
```

```
    }
}
```

---

25. Write a program to swap two numbers:
● With third variable
● Without third variable
-->

```
class swapNumbers{
    public static void main(String args[]) {
        int a = 5, b = 10;
        // With third variable
        int temp = a;
        a = b;
        b = temp;
        System.out.println("With temp: a = " + a + ", b = " + b);
        // Without third variable
        a = a + b;
        b = a - b;
        a = a - b;
        System.out.println("Without temp: a = " + a + ", b = " + b);
    }
}
```

---

26. Write a program to check whether a number is Armstrong.
-->

```
import java.util.Scanner;
class ArmstrongNumber {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        int sum = 0;
        int temp = n;
        while (temp > 0) {
            int d = temp % 10;
            sum += d * d * d;
            temp /= 10;
        }
        System.out.println(n + " is " + (sum == n ? "Armstrong" : "not Armstrong"));
    }
}
```

---

27. Take marks of 5 subjects and:
● Calculate total
● Calculate percentage
● Print grade (A/B/C/Fail)
-->

```
 import java.util.Scanner;
class MarksDemo {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        int total = 0;
        for (int i = 1; i <= 5; i++) {
            System.out.print("Enter marks " + i + ": ");
            total += sc.nextInt();
        }
        double percent = total / 5.0;
        String grade = percent >= 60 ? "A" : percent >= 40 ? "B" : percent >= 33 ? "C" : "Fail";
        System.out.println("Total: " + total + ", Percentage: " + percent + ", Grade: " + grade);
    }
}
```

---

28. Take user age and:
● Check eligibility for voting
-->

```
import java.util.Scanner;
class votingEligibilityDemo{
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter age: ");
        int age = sc.nextInt();
        System.out.println(age >= 18? "Eligible to vote" : "Not eligible to vote");
    }
}
```

---

29. Create a menu-driven program:
1. Add
2. Subtract
3. Exit

-->

```
import java.util.Scanner;
class MenuDrivenDemo {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        int choice;
        do {
            System.out.println("1. Add\n2. Subtract\n3. Exit");
            choice = sc.nextInt();
            if (choice == 1) {
                System.out.print("Enter two numbers: ");
                int a = sc.nextInt();
                int b = sc.nextInt();
                System.out.println("Sum: " + (a + b));
            } else if (choice == 2) {
                System.out.print("Enter two numbers: ");
                int a = sc.nextInt();
                int b = sc.nextInt();
                System.out.println("Difference: " + (a - b));
            }
        } while (choice != 3);
    }
}
```

---

30. Take a number and print its table until user stops (use loop).

-->

```
import java.util.Scanner;
class TableLoopDemo{
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        char ch;
        do {
            System.out.print("Enter a number: ");
            int n = sc.nextInt();
            for (int i = 1; i <= 10; i++) {
                System.out.println(n + " x " + i + " = " + (n * i));
            }
            System.out.print("Continue? (y/n): ");
            ch = sc.next().charAt(0);
        } while (ch == 'y');
    }
}
```

---

31. Fix the error:

```
int x;

-->
class FixErrorDemo{
    public static void main(String args[]) {
        int x = 0;                    // initialize x
        System.out.println(x);
    }
}
```

---

32. Fix:

```
if(x = 10){
System.out.println("Hello");
}
--> if(x == 10){                  //// use == for comparison, not =;
System.out.println("Hello");
}
```

33. Fix infinite loop:

```
while(i < 10){
System.out.println(i);
}
--> int i = 0;
    while (i < 10) {
    System.out.println(i);
    i++;  // to avoid infinite loop
}
```

---

34. Print all prime numbers between 1 to 100.

```
-->
class PrimeNumbersDemo {
    public static void main(String args[]) {
        for (int num = 2; num <= 100; num++) {
            boolean b = true;
            for (int i = 2; i <= Math.sqrt(num); i++) {
                if (num % i == 0) {
                    b = false;
```

```
                break;
            }
        }
        if (b) {
            System.out.println(num);
        }
    }
  }
}
```

---

35. Find GCD of two numbers.

```
-->class GCDdemo {
    public static void main(String[] args) {
        int a = 56, b = 98;
        while (b != 0) {
            int temp = b;
            b = a % b;
            a = temp;
        }
        System.out.println("GCD: " + a);
    }
}
```

---

36. Find LCM of two numbers.

```
-->class LCMDemo{
    public static void main(String[] args) {
        int a = 12, b = 15;
        int gcd = gcd(a, b);
        int lcm = (a * b) / gcd;
        System.out.println("LCM: " + lcm);
    }
    static int gcd(int x, int y) {
        if (y == 0) return x;
        return gcd(y, x % y);
    }
}
```

---

37. Check whether a number is perfect number.

```
--> class perfectNumberdemo{
    public static void main(String[] args) {
        int num = 90;
```

```
        int sum = 0;
        for (int i = 1; i < num; i++) {
            if (num % i == 0) {
                sum += i;
            }
        }
        if (sum == num) {
            System.out.println(num + " is a perfect number");
        } else {
            System.out.println(num + " is not a perfect number");
        }
    }
}
```

---

38. Create a number guessing game.

--> import java.util.Scanner;

```
import java.util.Random;

class GuessingGameDemo {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        Random rand = new Random();
        int numberToGuess = rand.nextInt(100) + 1;
        int userGuess = 0;

        while (userGuess != numberToGuess) {
            System.out.println("Guess a number between 1 and 100:");
            userGuess = sc.nextInt();
            if (userGuess > numberToGuess) {
                System.out.println("Too high!");
            } else if (userGuess < numberToGuess) {
                System.out.println("Too low!");
            } else {
                System.out.println("You guessed it!");
            }
        }
    }
}
```